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Machine Learning-based Automated System for Subjective Answer Evaluation

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***Abstract*—An examination is a useful tool for assessing students' knowledge. Evaluation of exams is a difficult and time-consuming process. The automatic examination of answer scripts makes this task easier for teachers, reducing the amount of effort and time required. The existing literature has a number of methods that have been proposed for evaluating responses to objective questions using machine learning. However, more work needs to be done on evaluating answers to descriptive questions. This study suggests a way to evaluate students' answers to questions of a descriptive kind without using traditional paper or pencil by teachers. Instead, a computer acts as a teacher and grades the students' submissions. The primary objective is to communicate the outcomes of subjective responses using Bidirectional Encoder Representations from Transformers (BERT), cosine, and Jaccard distance. The proposed model achieved an accuracy of 91%, an error of 9.01, a precision of 83%, and a recall of 79%, respectively. The suggested model has provided the best results in comparison with state-of-the-art systems.**

***Index Terms*—Answer Evaluation, BERT, Cosine Similarity, Jaccard Distance, Machine Learning**

I. Introduction

The standard work used in many areas in the assessment of students learning process is computer-based evaluation of student answers. The brilliant concept of incorporating computers into the learning process has revolutionized the world of education. The computer-assisted assessment method was created to analyze one-word answers, such as those found in multiple-choice questions. Also, based on keyword matching, it is possible to analyze paragraph answers, such as descriptive answers. The examination measures a student's learning abilities and is an essential aspect of education. After the examination, the teachers devote the majority of their time to reviewing the students' grades, which consumes a significant amount of human work, time, and money. An automated assessment evaluation system can help in saving time and effort during the process.

The evaluation of grades for a paper checker is easier if the questions are multiple-choice questions (MCQs). These questions are even easier to automate, and the system is already available in the market. However, automating the grading process for subjective questions is quite tedious and challenging. It is high-concentration work for teachers and requires time, effort, time, and cost. Few systems have been proposed to perform grading evaluations of descriptive questions [1], [2]. The subjective evaluation is based on the human understanding of a question and has high variation in grading. The existing systems evaluate the subjective questions based on the text, semantics, and spelling of the answer. To design an automated system, the evaluation must be capable of scoring the answer papers similarly to the human evaluators. There must be consistency in the grading pattern and should match the evaluator's grading. The current trend is solving the issues of grading systems using machine learning and deep learning methods. These methods have provided more promising results. There exist models capable of evaluating essays using bag of words, keyword matching, and sequence matching techniques.

The main aim of this work is to develop an automated system to grade subjective questions, reducing the evaluator's work. In this work, the automatic evaluation of the answers is performed using a model answer-based approach. The input given to the model is the question, teacher answer, and student answer. The proposed system initially does a feature extraction step. The answers are evaluated based on various features, which include sentence similarity with Bidirectional Encoder Representations from Transformers (BERT) between the model answer and the student's answer, cosine similarity, and Jaccard Distance. The input is fed to a pre-trained model. The BERT model takes three similarity scoring techniques, and the final prediction score is generated. These values are combined to generate the final score for the student's answer by taking a specific percentage of each evaluation method. The sigmoid activation function gives the final answer. If score is 0, answer is wrong, and if score is one, answer is correct.

The rest of the paper is organized as follows: Section II gives the literature review on the existing systems for grading subjective questions. Section III discussed preliminaries. Section IV provides the data collection procedure. Section V provides a detailed description of the proposed model. Section VI gives the detailed experimental analysis performed for the proposed approach. The conclusion is presented in section VII.

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TABLE I
LITERATURE REVIEW OF EXISTING SYSTEMS FOR MACHINE-LEARNING BASED ANSWER EVALUATION

Ref.	Dataset	Techniques	Performance	Remarks
[3]	Question-Answer dataset	Simple LSTM, Deep LSTM, Bi-directional LSTM	Accuracies = 83%,82%,89%	The obtained accuracy is commendable but the model was built on very few samples.
[4]	Question-Answer dataset	NLP- Sentiment Analysis, ANN	Accuracy = 90%	The model provided better accuracy but sentiment analysis tools do a really great job of analyzing text for opinion and attitude.
[5]	Sparse Dataset	Bayes Theorem	Accuracy = 80%	The method exhibited very good results. However, the drawback of this method is one must be able to convert the subjective prior beliefs into a mathematically stated prior.
[6]	Answers collected from different students to create the dataset	K-Nearest Neighbor	Accuracy = 76%	The method produced good results, but it is computationally expensive in terms of both time and storage.
[7]	Subjective question answers dataset	Word mover's distance, multinomial naive Bayes	Accuracy = 86%	The method provided good result, however the model was built on very few samples.
[8]	Kaggle dataset	Neural Network	(SSWE + Two-layer Bi-LSTM) Spearman's ρ = 0.91 Pearson r = 0.96 RMSE =2.4 Cohen's K = 0.96	Neural networks usually require more data, although there are some cases where neural networks do well with less data.
[9]	50 question answers collected from 10 students	Jaccard Similarity, Classification Algorithm	Precision = 95% Recall = 94% Sensitivity = 94.5%	The method exhibited very good results, but the model was built on very few samples.

II. LITERATURE REVIEW

A detailed survey of the existing systems to perform automated subjective question grading in the literature is presented in Table 1. The methods proposed for grading subjective questions can be attained from the current state-of-the-art systems by machine learning and deep learning methods. Even though the methods have resulted in promising performances, certain challenges still exist in the current systems. The main drawback of the current system is that it is evaluated on fewer datasets. The model may perform poorly for larger datasets. The similarity methods usually tend not to consider the semantic meaning of the statement. This can cause the method to categorize the wrong label, even if the answer is semantically correct. Therefore, automating the process of grading descriptive/subjective questions can be challenging task. However, the combination of Natural Language Processing (NLP) and deep learning methods can be the solution to the issues that exist in this problem.

III. PRELIMINARIES

1) BERT: It stands for Bidirectional Encoder Representations from Transformers. BERT is a free and open-source machine learning framework for dealing with NLP. BERT uses the surrounding text to provide the context to help computers understand the meaning of ambiguous words in the text. Using text from Wikipedia, the BERT framework is pre-trained, and question-and-answer datasets can be used to fine-tune it. The proposed model is inspired to employ the BERT, a contextualized embedding, for sentence similarity modeling in answer selection tasks. It can capture a better representation of a sentence by producing embedding of each word depending on its surrounding context. The working principle of BERT is illustrated in Figure 1.

We are using the Sentence Transformers package to access pre-trained models. As our use case needs functionality for English, we are using the bert-base-nli-mean-tokens pre-trained model. We should be to compare the similarity of sentences using something such as cosine similarity.

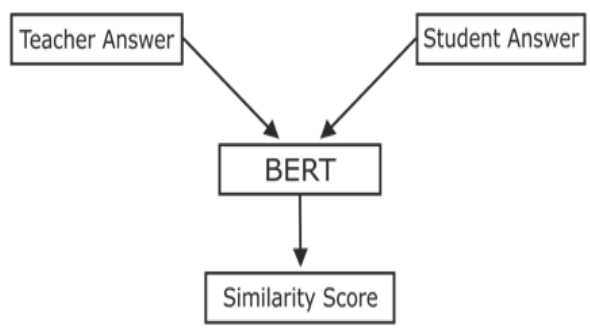

Fig. 1. Working principle of BERT

For the purposes of assessing Recall and Precision, each token in x is compared to the token $\hat{x}$ that is the closest match, and vice versa. The pairing is greedy and solitary. The F1 score is calculated by combining Precision and Recall. The metrics precision, recall, and F1 score can be computed as given in Equations 1, 2, and 3.

$$R_{BERT} = \frac{1}{|x|} \sum_{x_i \in x} \underset{\hat{x}_j \in \hat{x}}{max}\, x_i^T \hat{x}_j \tag{1}$$

$$P_{BERT} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in x} \underset{x_i \in x}{max}\, x_i^T \hat{x}_j \tag{2}$$

$$F_{BERT} = 2\frac{P_{BERT \cdot R_{BERT}}}{P_{BERT + R_{BERT}}} \tag{3}$$

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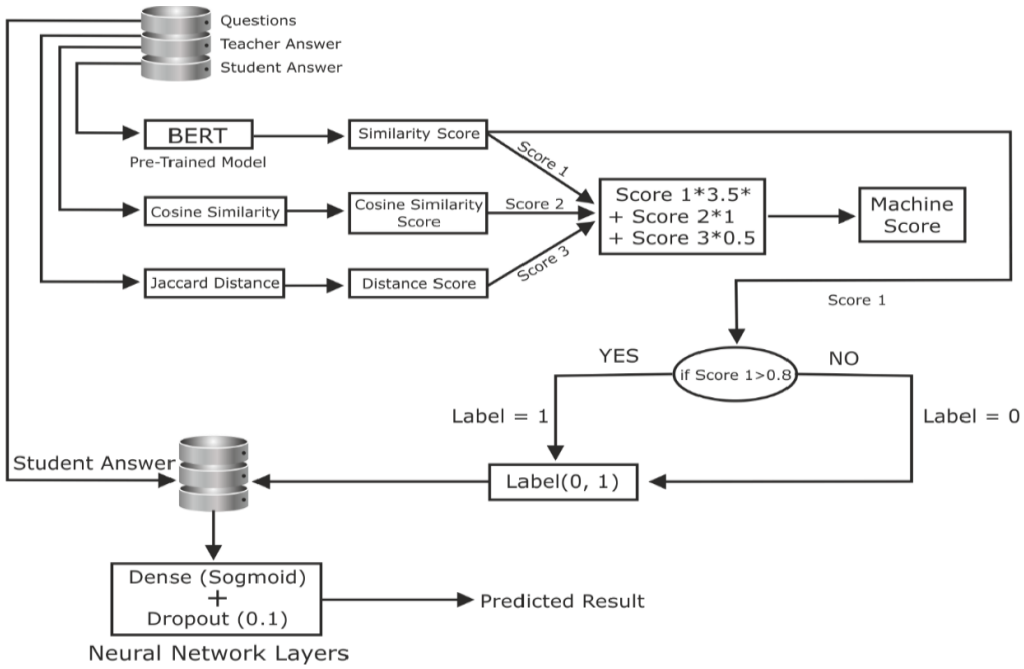

Fig. 2. Schematic architecture of the proposed system

2) Cosine Similarity : It is a formula that is used to check for text similarity, which is why it is needed in recommendation systems, question and answer systems, and plagiarism checkers. The fundamental idea behind cosine similarity is to measure the distance between two vector angles. The Euclidean dot product formula can be used to get the cosine of two non-zero vectors as shown in Equation 4:

$$Sim(A,B) = \frac{A \cdot B}{\|A\| \, X \, \|B\|} = \sum_{1}^{n} \frac{A_i \; X \; B_i}{\sqrt{\sum_1^n A_i^2 X} \sqrt{\sum_1^n B_i^2 X}} \tag{4}$$

3) Jaccard Distance: It measures the dissimilarity between sample sets, is a complement to the Jaccard coefficient, and can be calculated by deducting the Jaccard coefficient from 1 or dividing the union's size by the difference between the sizes of the intersection and union of two sets. This is defined as the ratio of the intersection of a set to the union of the set.

$$JaccardDistance = 1 - JaccardSimilarity$$
$$d_j = (A,B) = 1 - J(A,B) = \frac{|A \bigcup B| - |A \bigcap B|}{|A \bigcup B|} \tag{5}$$

IV. Data Collection

The data collection is performed by teachers and students. The process consisted of conducting the exam for five students, and the question paper contains 100 questions, and each question carries five marks. The teacher's answers for all 100 questions are recorded and the answers of the five students' answers for all 100 questions are also recorded. Thus, the dataset contains questions, teacher answers, and students' answers.

V. Proposed Methodology

Figure 2 shows the proposed system architecture for automatically evaluating descriptive answers. The method takes input from the database where the questions, answers of the teacher, and student answers are stored. Initially, the BERT model is pre-trained to obtain the similarity score of teacher and student answers. The other two similarity measures used are cosine similarity and Jaccard distance. To compute the final Machine score generated by the system, the scores generated from the three similarity measures are combined with a ratio of 3.5:1:0.5. The achieved Machine score is then considered as the final score returned by the system. This is the training phase of the system. In the testing phase, the student's answer is compared with the score generated by the system. The final label is given by the neural network consisting of a dense layer and a sigmoid activation function. The dropout layer is added to regularize the neural network to avoid overfitting. The final predicted result is the label 0 or 1, corresponding to the right or wrong answer. The parameters considered for training the proposed neural network are as follows: The optimization function used in Adam optimizer in which the learning rate parameter is set to 0.001. The model is trained for a number of epochs to achieve better performance.

The idea is to evaluate the descriptive answers from the model answer provided. The teacher provides a model answer based on which student's answer is assigned marks based on various parameters. For this evaluation the parameters used are BERT for sentence similarity, Cosine Similarity, and Jaccard Distance as shown in Figure 3.

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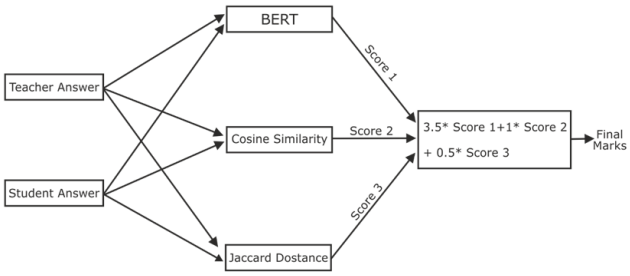

Fig. 3. Final Score Calculated using BERT, Cosine Similarity and Jaccard Distance

VI. Results and Discussion

In this section, the performance of the proposed model for automatic answer evaluation is discussed. For the evaluation, the answers of five students are considered with their corresponding human answers. The algorithm will calculate the similarity score and distance between human and student answers. All answers are evaluated firstly by the human then our algorithm will evaluate them. The score assigned by the computer and human evaluators is given in Table II.

BERT is used in this work because BERT is bidirectional, meaning it can handle text from both the left and right sides. BERT uses the encoder segment of the transformer model. Cosine similarity is a statistic for comparing the text-similarity of two texts, regardless of their size. A vector representation of a word is created. In n-dimensional vector space, the text documents are represented. The Jaccard distance is a measure of how dissimilar two sets are. BERT and Cosine are used to test several distance metrics, which found that the Jaccard distance performed better than the others. Also, when duplication does not matter, Jaccard similarity is advantageous. There are various applications for the simple, intuitive Jaccard Similarity formula, including object detection in image recognition, classification, and picture segmentation tasks (instance detection). Initially, BERT, cosine similarity score, and Jaccard distance are taken individually and the Machine score is calculated. However, the individual similarity metrics did not result in better score. Therefore, a Machine score was developed, which combines the properties of all the three measures and also result in improved performance. When BERT score and cosine score are compared, it was observed that BERT provided a better score than Cosine. Thus BERT score was multiplied by 3.5 and Cosine by 1. The Jaccard distance is a measure of how different two sets are. As a result, it produces a low score multiplied by 0.5. We experimented with several combinations of BERT, Cosine, and Jaccard values, but we discovered that the above-mentioned value produced the best results. As a result, the final BERT, Cosine, and Jaccard ratio is 3.5:1:0.5.

$$Machinescore = [(3.5*BERT)+(1*Cosinesimilarity) + (0.5*JaccardDistance)] \quad (6)$$

A. Machine Score and Human Score

The below figures show the comparison between Machine and Human scores of each answer for five students. For Question 1, a comparison of evaluation of results derived by the formula with that given by human values.

Question 1 - What are nodes? The Evaluation node guides how to quickly evaluate and link predictive models to select the most appropriate model for a particular application. Evaluation charts show how models perform in predicting specific outcomes. Figure 4 depicts the Machine and human

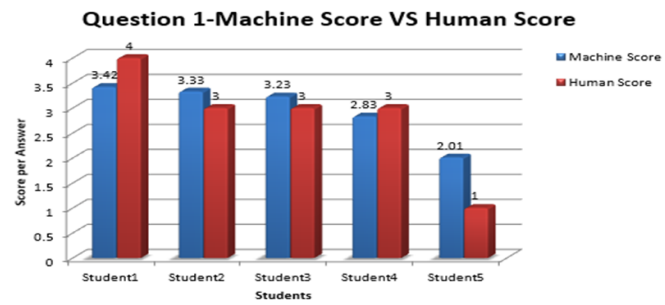

Fig. 4. Machine Score VS Human Score for Question 1

scores for question 1 for a group of five students. For student 1, Machine score is 3.42, and the human score is 4. For student 2, Machine score is 3.33, and the human score is 3. For student 3, Machine score is 3.23, and the human score is 3. For student 4, Machine score is 2.83, the human score is 3, and for student 5 Machine score is 2.01, and the human score is 1. We can observe that the difference between human and Machine scores is exceptionally modest.

Question 2 - What is the network? A method that emphasizes measuring the value of intercession as perceived by the (intended) beneficiaries, thereby pointing to their significance and apprehensions.

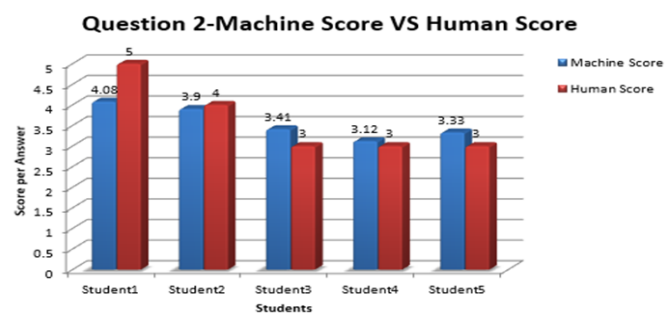

Fig. 5. Machine Score VS Human Score for Question 2

Figure 5 depicts the Machine and human scores for question 2 for a group of five students. For student 1, the Machine score is 4.08, and the human score is 5. For student 2, the Machine score is 3.9, and the human score is 4. For student 3 Machine score is 3.41, and the human score is 3. For student 4 Machine score is 3.12, and the human score is 3, and for student 5 Machine score is 3.33, and the human score is 3. We can see that the gap between human and Machine scores for question 2 is also relatively small.

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TABLE II
SCORE ASSIGNED BY THE MACHINE AND HUMAN

Question	Model Answer	Student Answer	BERT	Cosine similarity	Jaccard Distance	Machine Score	Human Score
Question 1	Teacher Answer	Student 1	0.74	0.67	0.32	3.42	4
		Student 2	0.75	0.57	0.27	3.33	3
		Student 3	0.79	0.33	0.27	3.23	3
		Student 4	0.72	0.26	0.1	2.83	3
		Student 5	0.5	0.11	0.3	2.01	1
Question 2	Teacher Answer	Student 1	0.88	0.84	0.33	4.08	5
		Student 2	0.94	0.49	0.25	3.90	4
		Student 3	0.81	0.40	0.35	3.41	3
		Student 4	0.76	0.37	0.19	3.12	3
		Student 5	0.81	0.33	0.33	3.33	3
Question 3	Teacher Answer	Student 1	0.80	0.33	0.25	3.28	3
		Student 2	0.81	0.12	0.19	3.07	3
		Student 3	0.81	1	0.15	3.91	4
		Student 4	0.66	0.13	0.26	2.64	2
		Student 5	1	0.04	0.25	3.66	4
Question 4	Teacher Answer	Student 1	0.54	0.81	0.28	2.84	3
		Student 2	0.96	0.64	0.26	4.13	4
		Student 3	0.59	0.45	0.47	2.75	2
		Student 4	0.78	0.35	0.19	3.175	3
		Student 5	0.88	0.20	0.21	3.385	3
Question 5	Teacher Answer	Student 1	0.80	0.79	0.36	4.43	5
		Student 2	0.83	0.49	0.40	3.595	4
		Student 3	0.76	0.48	0.43	3.355	3
		Student 4	0.91	0.43	0.34	3.785	4
		Student 5	0.78	0.27	0.45	3.225	2

Question 3 - What do you mean by network topology? The network topology describes such critical parameters as latency, ease of routing, throughput, and fault tolerance.

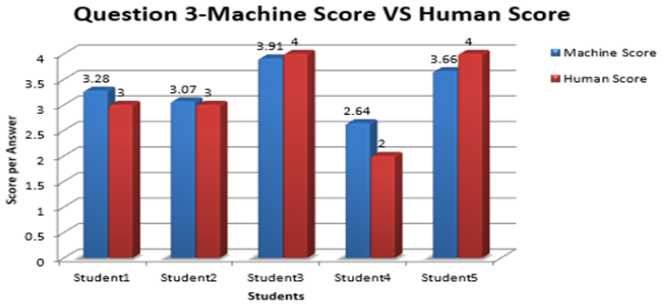

Fig. 6. Machine Score VS Human Score for Question 3

Figure 6 depicts the Machine and human scores for question 3 for a group of five students. For student 1, Machine score is 3.28, and the human score is 3. For student 2 Machine score is 3.07, and the human score is 3. For student 3 Machine score is 3.91, and the human score is 4. For student 4, the machine score is 2.64, the human score is 2, and for student 5, the machine score is 3.66, and the human score is 4. The difference between human and Machine scores for question 3 is also comparatively small.

Question 4 - What is bandwidth? The problem that typically arises at every company is network performance, e.g., low throughput and inefficient bandwidth utilization. It is the maximum amount of data transmitted over an internet connection in a given amount of time.

Figure 7 depicts the Machine and human scores for question 4 for a group of five students. For student 1, the Machine score is 2.84, and the human score is 3. For student 2 Machine score is 4.13, and the human score is 4. For student 3 Machine score is 2.75, and the human score is 2. For student 4 Machine score is 3.17, and the human score is 3, and for student 5 Machine score is 3.55, and the human score is 3. We can see that the difference between human and Machine scores for question 4 is also minimal.

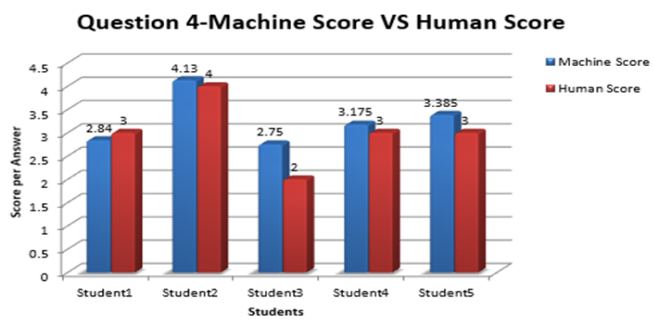

Fig. 7. Machine Score VS Human Score for Question 4

Question 5 - What are gateways? The primary function of the gateway is to provide a solo, reliable entry point for multiple APIs, irrespective of how they are implemented at the back end.

Figure 8 depicts the Machine and human scores for question 5 for a group of five students. For student 1, the Machine score is 4.43, and the human score is 4. For student 2 Machine score is 3.59, and the human score is 4. For student 3 Machine score is 3.35, and the human score is 3. For student 4 Machine score is 3.78, and the human score is 4, and for student 5 Machine score is 3.22, and the human score is 3. We can see that the difference between human and Machine scores for question 5 is also minimal.

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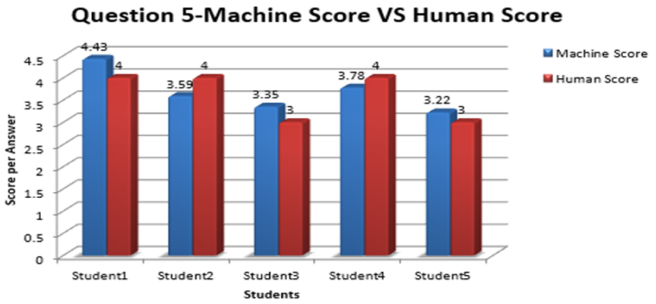

Fig. 8. Machine Score VS Human Score for Question 5

B. Error Rate

The error rate is calculated between Human and Machine score to evaluate the performance. Table III represents error between Machine and human score of each student.

TABLE III
ERROR BETWEEN MACHINE AND HUMAN SCORE

Students	Human Score	Machine score	Error
Student 1	181	185	4.00
Student 2	159	170.69	11.69
Student 3	152	165.91	8.91
Student 4	146	158	12.78
Student 5	155	164.56	9.56

C. Average Error Score

The average error is calculated to see which approach produces the lowest error. It has been observed that the Machine score, calculated using a formula, produces the lowest error when compared to BERT, Cosine, and Jaccard separately. The Average error score is given in Table IV. The table shows that BERT has a 32.79% higher average error score than the Machine score, Cosine has a 37.86% higher average error score than the Machine score, and Jaccard has an 82.61% higher average error score than the Machine score. Hence, the average error score proves that the Machine score that is calculated using the formula is suitable for calculating the score because it resulted in the least error.

TABLE IV
AVERAGE ERROR SCORE

Method	Average Error Score
Machine Score	9.01
BERT	41.8
Cosine	46.96
Jaccard	91.62

BERT is a bidirectional language representation model pre-trained with deep Bidirectional Transformers. In the proposed model, BERT layers with Neural Network layers are used. A dense layer is added with ReLU and sigmoid activation function. Adam optimizer with a learning rate of 0.001 is used. Below Table V shows the testing accuracy, loss, precision, and recall on the different epochs. The model evaluation result proves that neural networks are suitable for making predictions. The best accuracy of 91% was obtained for the BERT model with the Neural Network layer at 1000 epochs.

Table VI shows the comparison of previous works with the proposed work. The methods include machine learning methods such as Bayes Theorem, K-Nearest Neighbour, and multinomial naive Bayes, and deep learning algorithm like Long Short Term Memory (LSTM) neural network.

TABLE V
TESTING RESULT ON DIFFERENT EPOCH

Epoch	Loss	Accuracy	Precision	Recall
100	0.83	0.79	0.73	0.69
300	0.69	0.83	0.79	0.72
500	0.65	0.88	0.81	0.75
1000	0.42	0.91	0.83	0.79
2000	0.51	0.86	0.79	0.73
3000	0.49	0.82	0.81	0.68

TABLE VI
COMPARISON OF PREVIOUS WORKS WITH THE PROPOSED WORK

Ref.	Method	Performance
[3]	LSTM	Accuracy= 83%
[5]	Bayes Theorem	Accuracy= 80%
[5]	K-Nearest Neighbor	Accuracy= 76%
[7]	Word mover's distance, multinomial naive bayes	Accuracy= 86%
Proposed method	BERT, Cosine Similarity, Jaccard Distance	Accuracy= 91%, Error= 9.01, Precision= 83%, and Recall= 79%

VII. CONCLUSION

Scoring student responses is more difficult under the existing manual evaluation approach. The valuation scheme, which involves spending a lot of money, time, and human effort, raises further problems. An automatic evaluation system is suggested for evaluating answers of the descriptive type to overcome these challenges. The proposed system is an automated descriptive answer checking and scoring using BERT. The 100 different answer scripts that made up the dataset for the evaluation were manually assembled. Our approach obtained the best accuracy of 91%, precision of 83%, and a recall of 79%.

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